PROJECT 01

AI Lead Generation & CRM Automation

Turn incoming enquiries into organized, sales-ready leads.

THE PROBLEM

Teams lose time reading enquiries, qualifying prospects and updating CRM records manually.

WHAT THE DEMO DOES

- Capture lead details through an API
- Apply transparent lead-scoring rules
- Store records in SQLite (demo)
- Generate follow-up email drafts for human review

WHO IT HELPS

Marketing agencies • Recruiters • B2B service teams

BUSINESS VALUE

Less repetitive data entry; faster, more consistent lead triage.

TECH STACK

Python • FastAPI • Pydantic • SQLite • REST APIs

NEXT STEP Add n8n webhook, CRM sandbox integration, duplicate detection and secure authentication.

PROJECT 02

AI Customer Support Draft Assistant

Classify support requests and draft policy-aware replies.

THE PROBLEM

Support teams repeatedly answer common questions and need to route sensitive issues to people.

WHAT THE DEMO DOES

- Classify incoming tickets
- Retrieve relevant sample policy content
- Draft a customer reply
- Escalate urgent or unmatched cases for human review

WHO IT HELPS

E-commerce • SaaS • Online service businesses

BUSINESS VALUE

Faster first drafts while keeping a human in control of customer responses.

TECH STACK

Python • FastAPI • JSON knowledge base • Pydantic

NEXT STEP Connect a helpdesk sandbox, add retrieval evaluation, access controls and approved-policy ingestion.

PROJECT 03

AI Document Processing & Validation

Convert document data into structured records with validation checks.

THE PROBLEM

Invoice and operations teams manually enter fields and check whether totals reconcile.

WHAT THE DEMO DOES

- Accept structured invoice fields
- Validate line-item subtotals
- Check subtotal plus tax against total
- Flag mismatches for review

WHO IT HELPS

Finance operations • Procurement teams • Back-office service providers

BUSINESS VALUE

A clear validation step before data enters downstream business systems.

TECH STACK

Python • FastAPI • Pydantic • Decimal validation

NEXT STEP Add PDF/OCR extraction, duplicate detection, confidence scores and secure review queues.

PROJECT 04

Business Knowledge Retrieval Assistant

Find relevant answers in a collection of company documents.

THE PROBLEM

Employees spend time searching policies and internal documentation for specific information.

WHAT THE DEMO DOES

- Index a small sample document set
- Rank passages using TF-IDF retrieval
- Return relevant excerpts and source IDs
- Indicate when no relevant passage is found

WHO IT HELPS

Consultancies • HR teams • Internal operations teams

BUSINESS VALUE

A transparent retrieval baseline with traceable source excerpts.

TECH STACK

Python • FastAPI • scikit-learn • TF-IDF

NEXT STEP Replace TF-IDF with embeddings/vector search; add permissions, citations and retrieval evaluation.

PROJECT 05

Safe Natural-Language Data Analyst

Ask business questions and get answers from sample sales data.

THE PROBLEM

Non-technical users need quick summaries without writing SQL themselves.

WHAT THE DEMO DOES

- Recognize a small set of supported questions
- Run predefined read-only SQL queries
- Summarize revenue by product, region or quarter
- Return structured results with clear limitations

WHO IT HELPS

Small businesses • Sales teams • Analytics consultancies

BUSINESS VALUE

Demonstrates a safer starting point than executing arbitrary model-generated SQL.

TECH STACK

Python • FastAPI • SQLite • SQL

NEXT STEP Add schema-aware query planning, validation, role-based access, query limits and dashboard charts.

PROJECT 06

Source-Traceable Research Report Assistant

Turn supplied research notes into a structured, reviewable report draft.

THE PROBLEM

Research and consulting workflows need organized evidence, source tracking and clear gaps.

WHAT THE DEMO DOES

- Accept a research question and source notes
- Retain source IDs and URLs
- Assemble a Markdown report draft
- Explicitly flag verification gaps and limitations

WHO IT HELPS

Consultants • Business development teams • Research agencies

BUSINESS VALUE

A traceable report scaffold that avoids pretending unverified sources were checked.

TECH STACK

Python • FastAPI • Pydantic • Source metadata

NEXT STEP Integrate an approved search provider, store timestamps, verify citations and add human review.